An Evaluation of Medications Commonly Used for Epidural Neurolysis Procedures in a Human Fibroblast Cell Culture Model

Christof Birkenmaier, MD, Julia Redeker, Birte Sievers, PhD, Carolin Melcher, MD, Volkmar Jansson, MD, and Susanne Mayer-Wagner, MD, PhD

His argument is contradictory

Background and Objectives: Epidural injections are popular therapies for sciatica and low back pain. Local anesthetics and corticosteroids are commonly used for most injections techniques, but some treatments use a specific combination of several agents. The epidural lysis of adhesions procedure (Racz) uses a combination of bupivacaine, hyaluronidase, a corticosteroid, and hypertonic saline. Because severe complications, some with permanent neurologic deficits, have been observed, we considered the possibility that individual agents or a combination thereof might be capable of damaging or destroying cells in primarily the epidural tissues.

Methods: We used monolayer cell cultures of human fibroblasts in Dulbecco modified Eagle medium to study these pharmacological agents alone or in combination. Cell viability and proliferation were assessed by Trypan blue staining, cell counts, and the WST-1 assay. Time and concentration series were performed.

Results: With the corticosteroid, we observed the previously described proliferation-retarding effects. Hyaluronidase was not found to have a relevant effect on fibroblast proliferation. Bupivacaine and hypertonic saline were found to have a time- and concentration-dependent effect on cell viability and proliferation. Both were found to be toxic at concentrations well below the ones used clinically.

Conclusions: We identified a potential for harm caused by commonly used pharmacological agents when applied epidurally. Animal studies will have to show whether the same can be observed in living tissues.

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There are a large number of minimally invasive treatments for low back pain and sciatica, many of them without sufficient evidence to support their use. Epidural applications of medications make up a considerable portion of these therapies and epidural catheter techniques represent a part of this spectrum. One of them, the so-called "epidural lysis of adhesions and spinal injection via catheter" (also called epidural neurolysis; hereafter, LOA)[1] has been popular as an interventional therapy for sciatica with and without low back pain, caused by postsurgical scarring or by epidural adhesions of other origin. Beyond a purported mechanical mode of action, it uses a cocktail of 4 pharmacological agents: bupivacaine, triamcinolone acetonide, hyaluronidase, and hypertonic saline, which is different from other types of epidural injections that usually use only a local anesthetic and a steroid. Although there are some data supporting the efficacy of this treatment,[2–4] severe complications, some resulting in permanent cauda equina syndrome, have been observed. Some of these cases have been tried in federal court,[5] and the German National Medical Board, in its health technology assessment, categorized the LOA procedure as being experimental because of the absence of prospective and well-controlled studies showing the efficacy of this treatment.[6] As a consequence, some malpractice insurance companies now levy a risk premium for doctors who perform the LOA procedure. At our institution, we indicate for LOA only in cases of clear radiculopathy without a fresh motor deficit and with a matching anatomic correlate on magnetic resonance imaging (MRI; disk herniation, bulging, and postinflammatory or postsurgical scar). In our own experience of 173 procedures, 1 case of permanent urinary incontinence and 1 case of transient erectile dysfunction have occurred. Other complications included 2 sheared-off catheters that could be removed under local anesthesia and 4 recognized subdural catheter placements, which led to the abandoning of the procedure. The evaluation of the 2 incidents with a neurologic deficit included a thorough review of the epidurograms that were obtained and documented during the procedures and immediate MRI. Neither the epidurograms nor the MRI studies gave any indication that an unintentional subdural catheter placement and hence an intrathecal injection had occurred. We therefore judged it unlikely that technical errors were the reason for the neurologic deficits observed and hypothesized and that the unique combination of pharmacological agents used in this therapy might be responsible for some of the complications. Of the 4 agents used for the LOA procedure, only bupivacaine has been studied for and is approved for epidural use. Bupivacaine, in turn, has been shown to have toxic or inhibitory effects on chondrocytes, fibroblasts, muscle, melanoma, renal, and neural cells.[7–14] We decided to use a cell culture model to investigate whether bupivacaine, triamcinolone acetonide, hyaluronidase, and hypertonic saline combined or individually have the potential to damage or destroy cells. Because the epidural space is predominantly occupied by connective tissue, we chose a human fibroblast cell line for our in vitro experiments.

METHODS

Cell Culture

Human fibroblasts (lot no. 051H, Provitro, Berlin, Germany) were seeded in triple flasks (Nunc, Roskilde, Denmark) and grown at 37°C in a humidified atmosphere containing 95% air and 5% CO_2 in Dulbecco modified Eagle medium (DMEM;

From the Department of Orthopedic Surgery, Grosshadern Medical Center, University of Munich, Marchioninistr., Munich, Germany.
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Address correspondence to: Christof Birkenmaier, MD, Department of Orthopedic Surgery, Grosshadern Medical Center, University of Munich, Marchioninistr. 15, 81377 Munich, Germany (e-mail: christof.birkenmaier@med.uni-muenchen.de).
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Biochrom, Berlin, Germany) supplemented with 10% fetal bovine serum (PAA, Pasching, Austria) and 50 U/mL penicillin/streptomycin (Biochrom).

Preincubation

Cells were then retrieved from the triple flasks, rinsed, and plated at 10^4 cells per well on 6-well plates or on 96-well plates, depending on the individual experiment or assay to be performed. Subsequently, the cells were incubated in DMEM under standard conditions during 24 hrs to allow for adhesion.

Incubation

After preincubation, the DMEM was removed, the cells were gently rinsed with phosphate-buffered saline (PBS, pH = 7.4; Biochrom-Seromed). Then, DMEM containing either 1 of the 4 medications alone or a combination of all 4 agents at varying concentrations was added as cell culture medium. These agents were added to the DMEM as crystalline salts or powders to not reduce the concentrations of the original components of the DMEM. The enriched DMEM preparations were then sterile filtered. Incubation periods are as outlined for the individual experiments.

Cell Counts/Viability Testing

All cell counting procedures were performed using a hemocytometer in the presence of Trypan blue, all counts were repeated 3 times, and the mean was determined.

Cell Proliferation Assay

Cell proliferation was assessed with 4-[3-(4-Iodophenyl)-2-(4-nitrophenyl)-2*H*-5tetrazolio]-1.3-benzendisulfonate (WST-1-assay). WST-1 is a colorimetric assay based on the cleavage of the tetrazolium salt WST-1 by mitochondrial dehydrogenases in viable cells (Roche, Mannheim, Germany). At the measuring points of 0, 3, 6, 12, and 24 hrs, WST-1 solution (10 μL/well) was added to the 96-well plates, and the cells were incubated for 2 hrs at 37°C. The plate was read on an Elisa-Reader (MWG-Biotech, Ebersberg, Germany) at 450 nm, with a reference wavelength at 690 nm.

Time Response Incubation With Bupivacaine, Hyaluronidase, Triamcinolone Acetonide, and Hypertonic Saline

As a pilot experiment, we performed incubations in which the fibroblasts were exposed for 1 hr to DMEM containing all 4 reagents as a combination or individually in the same concentrations/dosages as used in the clinical application of these substances: bupivacaine 0.25%, NaCl 10%, triamcinolone at 40 mg/10 mL, and hyaluronidase at 1500 IU/10 mL. After 1 hr, the culture medium was removed, cells were gently rinsed with PBS, and pure DMEM was added as a medium. The cells were then cultured for up to 5 days, and cell counts were performed daily (Fig. 1). The results from this pilot experiment were used to define which agents were to be tested in details and which concentration range/incubation times to investigate.

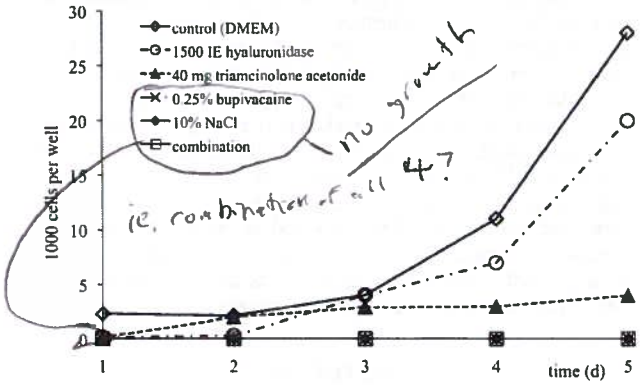

FIGURE 1. Effects of incubation of a human fibroblast monolayer culture for 1 hr with DMEM containing 0.25% bupivacaine, 10% sodium chloride, 1500 IU of hyaluronidase, and 40 mg of triamcinolone acetonide either as single substances or as a combination. Cell counts were performed with a hemocytometer at days 1 to 5 after incubation.

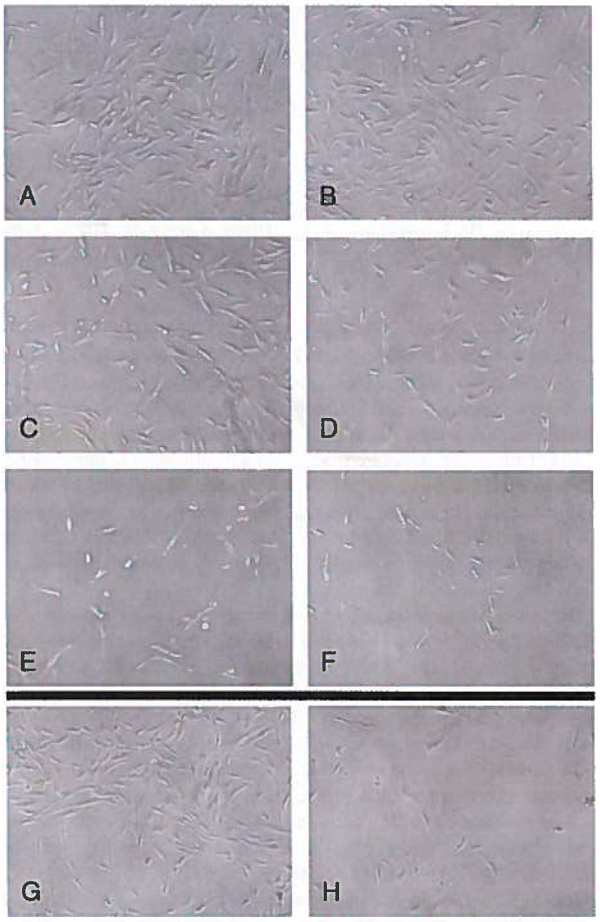

FIGURE 2. Microscopic photographs showing the dose-dependent reduction of viability of monolayer cultures from human fibroblasts on day 3 after 1 hr of incubation with DMEM as control group (A), 0.9% NaCl-DMEM (B), 2% NaCl-DMEM (C), 4% NaCl-DMEM (D), 6% NaCl-DMEM (E), and 10% NaCl-DMEM (F). Below the black bar are representative pictures on day 3 after 24 hrs of incubation with DMEM as control group (G) and 0.9% NaCl-DMEM (H).

Dose/Time–Response Incubations With Hypertonic Saline

Cells were plated at 10^4 cells per well on 6-well plates and preincubated as indicated above. The DMEM was then removed, and cells were rinsed with PBS. Hypertonic DMEM (supplemented with 0.9%, 2%, 4%, 6%, and 10% NaCl) was prepared from DMEM and NaCl (Merck, Darmstadt, Germany), and cells were incubated with the hypertonic culture medium. After 1 hr of treatment, cells were rinsed twice with PBS and cultured with

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DMEM for 1 to 3 days. Cells were photographed after treatment (Fig. 2). Total cell count was performed with a hemocytometer (Neubauer chamber). Cells were counted on days 1, 2, and 3 after treatment. Each well was counted 3 times, and the mean was determined (Fig. 3).

Dose/Time–Response Incubations With Bupivacaine

Cell culture and adhesion were performed as outlined above. Cells were suspended in a treatment medium of DMEM containing 0.1%, 0.01%, and 0.05% of bupivacaine (AstraZeneca, Wedel, Germany) and plated in 96-well plates at 10^4 cells per well under standard conditions for 1 hr. After incubation, the 96-well plates were centrifuged, the bupivacaine-containing culture medium was removed, and cells were gently rinsed with PBS (pH 7.4). Dulbecco modified Eagle medium was added, and cells were incubated under standard conditions as indicated above for 24 hrs. Assessment of cell proliferation was performed after 0, 3, 6, 12, and 24 hrs (Fig. 4).

Statistical Analysis

Results are presented as the mean ± SD. All experiments were repeated at least twice, unless otherwise specified. Statistical analysis was performed using Student *t* test.

RESULTS

Pilot Experiment

Incubations for 1 hr with DMEM containing 10% saline—alone or in combination with other agents—led to the complete death of the cell culture. Incubations with hyaluronidase alone showed only a minimal effect when compared with control, which was not significant. Incubations with triamcinolone acetonide alone showed the proliferation-retarding effect that has long been known.[15,16] Incubations with 0.25% bupivacaine alone and in combination also led to the complete death of the cell culture (all shown in Fig. 1). So in essence, the proliferation-retarding effect of the corticosteroid was confirmed, and hyaluronidase was found to have no relevant effect on the cell culture. This prompted us to further investigate the effects of saline and bupivacaine in different concentrations and for different incubation times.

Incubation With Hypertonic NaCl

The effects of 1-hr incubation with different NaCl-DMEMs are shown in Figures 2 and 3. On days 1 and 2, cell viabilities after 1 hr of incubation with 2% NaCl-DMEM, 4% NaCl-DMEM, 6% NaCl-DMEM, and 10% NaCl-DMEM were significantly reduced compared with control ($P < 0.05$). On day 3, cell viabilities in all groups were significantly reduced compared with control ($P < 0.05$) with the effect of 0.9% NaCl-DMEM and 2% NaCl-DMEM being much less than with the stronger concentrations. There were no remaining viable cells with 10% NaCl-DMEM. Figure 3 shows the dose-dependent reduction in viability, and Figure 2 is composed of representative microscopic pictures. When 24-hr incubations with NaCl-DMEMs were performed, only the 0.9% NaCl-DMEM showed cell survival beyond days 1 and 2. The cell counts observed on day 3 after 24 hrs of incubation with 0.9% NaCl-DMEM were similar to those with 6% NaCl-DMEMs or higher after 1 hr of incubation (Fig. 2).

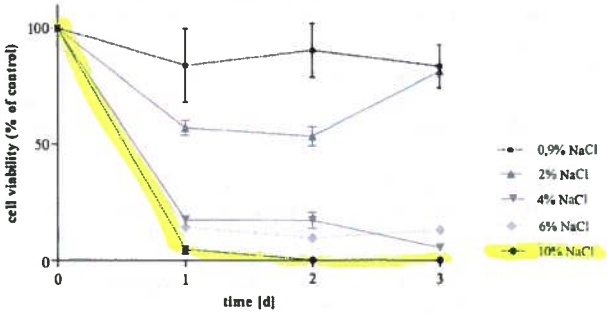

FIGURE 3. Effects of 1 hr of incubation with DMEM containing different NaCl concentrations on human fibroblasts. Cell counts were performed with a hemocytometer on days 1, 2, and 3. The graph shows cell numbers normalized to the control group during a 3-day period. There was a reduction in cell viability with increasing NaCl concentrations.

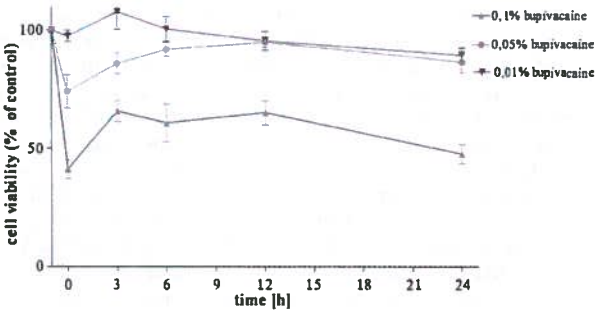

FIGURE 4. Effects of 1 hr of incubation with different bupivacaine-DMEMs on human fibroblasts. Cell proliferation was assessed using the WST-1 assay after 0, 3, 6, 12, and 24 hrs and normalized to control. Cell viabilities decreased initially for bupivacaine 0.1% and 0.05% but not for 0.01%. After 12 hrs, only bupivacaine 0.1% showed viabilities that were significantly reduced, beyond 12 hrs, a decline in viability compared with control was observed in all groups. Cell survival decreased with increasing bupivacaine concentrations, whereas the group with 0.01% of bupivacaine-DMEM and the group with 0.05% of bupivacaine-DMEM showed similar cell survival.

Incubation With Bupivacaine

The WST-1 assay showed that bupivacaine in the cell culture medium caused a concentration-dependent reduction in fibroblast proliferation (Fig. 4). Immediately after incubation, the group with 0.01% bupivacaine-DMEM showed no significant difference compared with the control group. The group with 0.1% bupivacaine-DMEM and the group with 0.05% bupivacaine-DMEM showed significantly lower cell counts than the control group ($P < 0.05$). After 3 hrs and after 6 hrs, cell proliferation in the groups with 0.1% bupivacaine-DMEM and with 0.05% bupivacaine-DMEM was significantly less than that of the control ($P < 0.05$). After 12 hrs, only the group with 0.1% bupivacaine-DMEM exhibited proliferation significantly less than that of the control ($P < 0.05$). After 24 hrs, all groups were significantly different from control ($P < 0.05$) with the 0.05% bupivacaine-DMEM group having a viability of less than 50% of the control.

DISCUSSION

The LOA procedure is primarily used as an interventional therapy to treat sciatica with and without associated low back pain, thought to arise from epidural scarring or adhesions that are either caused by previous surgery or by inflammatory processes after annular rupture and/or disk herniation.

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Although hyaluronidase and hypertonic saline are almost exclusively used for this specific procedure, corticosteroid preparations of some sort are—despite conflicting evidence with regards to their added benefit over local anesthetics alone—a central component of many epidural injection techniques. An unsurprising finding of our initial pilot experiment was the proliferation-retarding effect of the corticosteroid, which is well established.[15,16] There has been discussion in the past about the toxic adverse effects of corticosteroid preparations with intrathecal administration, but these have been largely attributed to preservatives and additives of the respective preparations.[17] Hyaluronidases are found in many animal venoms, and they aid the penetration of these venoms into tissues. Therapeutically, these enzymes have been used for increasing the penetration of cytotoxic drugs into tumors, the reduction of skin necroses in extravasation injuries,[18] the penetration of corticosteroids into target tissues,[19] and the treatment of noninfectious spinal arachnoiditis.[20] There are only very limited data on formal toxicity studies that are not in the direct context of studying animal venoms. It seems reasonably clear, however, that any direct toxicity that had been suspected to be associated with this enzyme was related to impurities and additives of a specific preparation.[21,22] We found no significant cytotoxic effect of hyaluronidase when compared with control. Still, because this enzyme cleaves the hyaluronic acid molecules of the extracellular matrix, the absence of an observable effect in our experiments might be due to the inherent limitation of a cell culture experiment and may require further investigation in live tissue experiments. On the other hand, there are only few studies reporting on its use in epidural pain treatments; none of which is a true controlled trial.[23–26] The only study comparing the use of saline plus hyaluronidase to saline alone did not show a significant difference in the clinical effect, so there may simply be no need for this enzyme to be used in epidural injections.[26] With regard to hypertonic saline, there have been studies on the use of hyperosmotic solutions in cryoprotection for transplant surgery and other applications that have also looked at hypertonic sodium chloride solutions.[27–29] The earliest study[27] found damage to frog sciatic nerve axons when exposed to hypertonic salt solutions. In a more recent study,[29] the high chloride concentration of the solution was specifically found to be the main reason for cell toxicity in a human endothelial cell monolayer culture. The motility of canine spermatocytes[28] has been reduced by exposure to hypertonic saline as has been the ciliary activity of cryopreserved human nasal mucosa.[30] Hypertonic saline has also been shown to be toxic to cultures of myocytes and hippocampal neurons at different concentrations.[31,32] Given the fact that 10% NaCl is injected into the epidural space during the LOA procedure, it is noteworthy that the fibroblasts in our experiments did not tolerate hypertonicity well at all. Already the slight hypertonicity created by raising the NaCl concentration in the DMEM by 2% (from 0.64% to 2.64%) for a period of 1 hr led to a ~50% decrease in cell viability, whereas higher concentrations led to the immediate or delayed death of the cell culture. Unintentional injection of hypertonic saline has been implicated in neurologic complications,[33] and the questionable effect of hypertonic over isotonic saline injections has been critically discussed.[26,34] A controlled study found no difference whether hypertonic or isotonic saline was used for LOA.[3] Pending confirmatory experiments in a suitable animal model, we currently see no justification for using hypertonic saline epidurally for the lack of evidence regarding its effectiveness and because of our findings, which point at potential risks. We were surprised at the strong negative effect on fibroblast viability that we observed with even low concentrations of bupivacaine after an incubation of 1 hr. Because the washout of contrast from the epidural space occurs often after hours and because bupivacaine is mainly metabolized in the liver, the assumption of a local presence for 1 hr in the epidural space does not seem unreasonable. It is therefore conceivable that bupivacaine may have a clinically relevant cytotoxic effect on epidural tissues. Our study has the limitations of a cell culture experiment, which can be very sensitive to even minimal changes in the concentrations of electrolytes, tonicity, buffer substances, nutrients, and experimental agents. Our findings can therefore not be directly transferred to the situation in a living organism; however, we believe that the effects that we observed are a reason for caution and for further studies. In the past, cell culture experiments have been important in the studying and characterizing the toxic effects of lidocaine.[35,36] Three-dimensional cell culture experiments and animal experiments will have to show whether our findings are reproducible in more complex cultures and in live tissues. We did not study ropivacaine in this series of experiments because it is not part of the typical combination of medications for epidural LOA. Some data by other researchers, however, indicate that it might be less toxic than bupivacaine, and it may therefore be an alternative for epidural treatments.[9,37,38]

It remains a point of discussion whether fibroblasts were the appropriate choice of cell type for this research. Alternatively, cells of neural origin could have been used. However, the complications with the LOA procedure that we are aware of did not occur after unintentional subdural catheter placement, as was demonstrated by means of epidurograms. Because the epidural space is predominantly occupied by connective tissue and because the dura mater is also mostly composed of connective tissue, we decided on a human fibroblast cell line for our in vitro experiments. Still, some of the undesired effects observed with the LOA procedure could have resulted from a passage of the pharmacological agents through the dura and hence would require confirmation with a neural cell culture.

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